5 CREATE RELEVANT PATIENT SERIES

The antigen Supporting Data defines one or more antigen series for each antigen. Before beginning the evaluation process for a given patient, a set of relevant patient series must first be selected and created for the patient. Not all antigen series will be relevant for a given patient and only antigen series appropriate for the patient should be evaluated, forecasted, and considered for best patient series selection. The appropriateness of an antigen series is based on criteria such as patient gender, age, and underlying conditions.

TABLE 5-1 CREATE RELEVANT PATIENT SERIES PROCESS STEPS

Section	Activity	Goal
5.1	Select Relevant Patient Series	The goal of this step is to identify the antigen series which are appropriate for the patient.

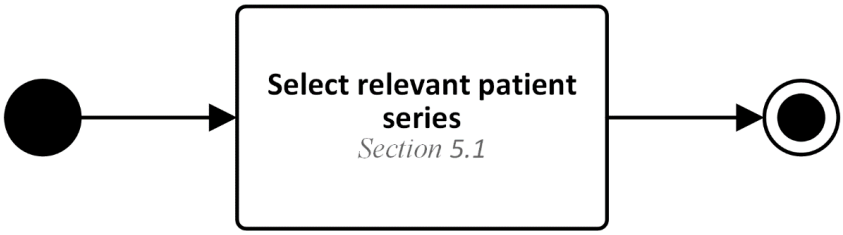

FIGURE 5-1 CREATE RELEVANT SERIES PROCESS MODEL

5.1 SELECT RELEVANT PATIENT SERIES

Select relevant patient series determines which series defined by the Supporting Data are appropriate to evaluate for the patient. Antigen series with a Series Type of “Standard” or “Evaluation Only” are relevant for all patients of the appropriate gender. Not all antigen series with a Series Type of “Risk” will be appropriate for a given patient.

Given the complex nature of indications, it may not always be possible to conclusively determine if an indication applies to a patient. To minimize false positive forecasts, in the case where a Risk Series cannot be definitively determined to be relevant for a patient (that is some or all indications are inconclusive and none unambiguously apply to the patient) the series will not be evaluated or forecast, but a notification should be available to a clinician alerting them to the presence of the indication(s) which could not be resolved.

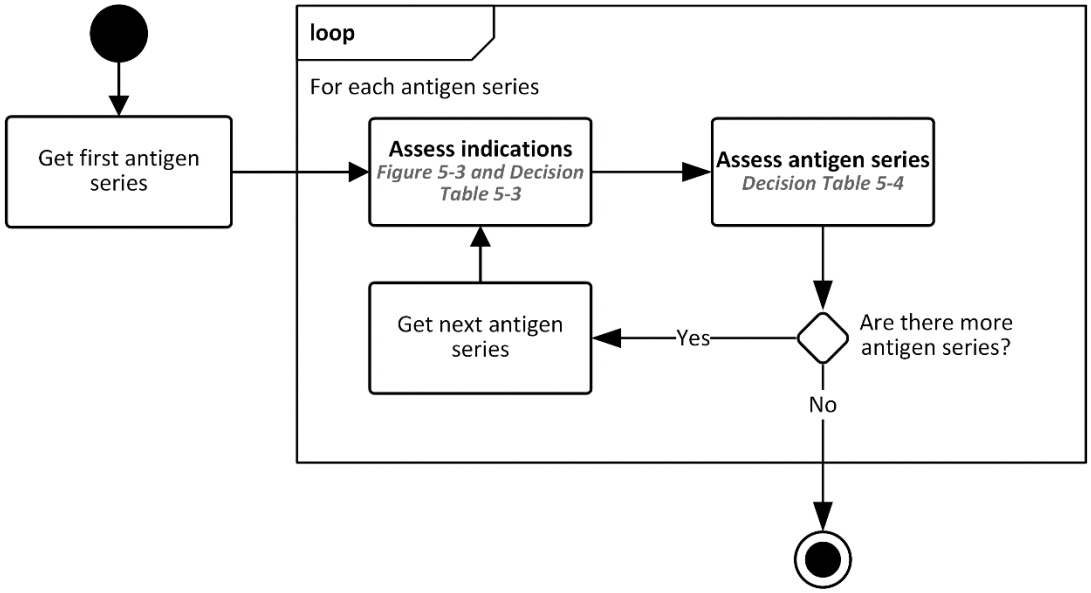

FIGURE 5-2 SELECT RELEVANT PATIENT SERIES PROCESS MODEL

TABLE 5-2 SELECT RELEVANT PATIENT SERIES ATTRIBUTES

Attribute Type	Attribute Name	Assumed Value if Empty
Patient	Gender	Unknown
Patient	Date of Birth	-
Patient history	Active Patient Observation(s)	-
Supporting Data (Gender)	Required Gender	Gender of the patient
Supporting Data (Series Type)	Series type	-
Supporting Data (Indication)	Observation Code	-
Runtime data	Assessment Date	current date
Calculated date (CALCDTIND-1)	Indication Begin Age Date	01/01/1900
Calculated date (CALCDTIND-2)	Indication End Age Date	12/31/2999

TABLE 5-3 SELECT RELEVANT PATIENT BUSINESS RULES

Business Rule ID	Business Rule
CALCDTIND-1	A patient's indication begin age date must be calculated as the patient's date of birth plus the indication begin age of an indication.
CALCDTIND-2	A patient's indication end age date must be calculated as the patient's date of birth plus the indication end age of an indication.

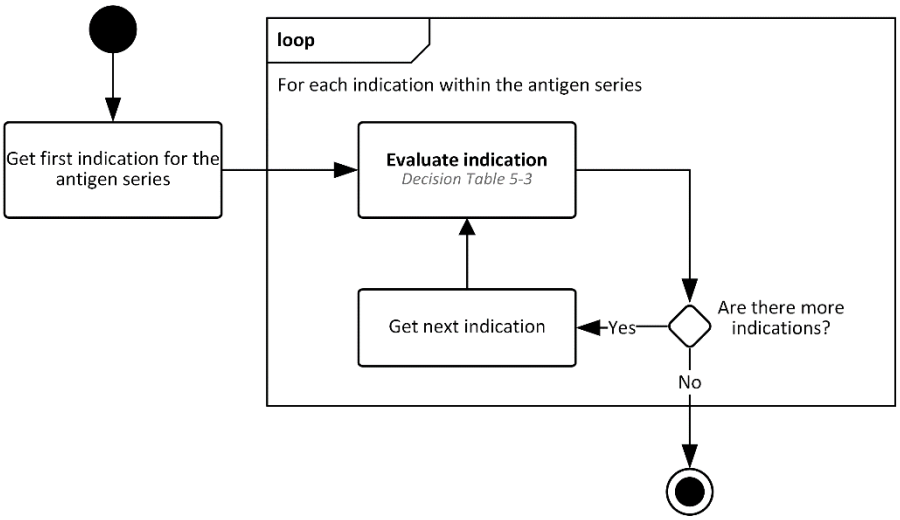

FIGURE 5-3 ASSESS INDICATIONS PROCESS MODEL

TABLE 5-4 DOES THE INDICATION APPLY TO THE PATIENT?

CONDITIONS	RULES			
Does the indication describe any active patient observations?	Yes	No	Unknown	-
Is the indication begin age date ≤ assessment date < indication end age date?	Yes	Yes	Yes	No
OUTCOMES	Yes. The Indication applies to the patient.	No. The Indication does not apply to the patient.	No. The Indication does not apply to the patient; however, the Indication Text Description should be made available to the clinician for manual determination.	No. The indication does not apply to the patient.

TABLE 5-5 IS AN ANTIGEN SERIES A RELEVANT PATIENT SERIES FOR A PATIENT?

CONDITIONS	RULES			
Is the patient gender one of the required genders of the antigen series?	Yes	No	Yes	Yes
Is the series type of the antigen series 'Standard' or 'Evaluation Only'?	Yes	-	No	No
Does at least one indication that drives the need for the antigen series apply to the patient?	-	-	Yes	No

CONDITIONS	RULES			
OUTCOMES	Yes. The antigen series is a relevant patient series for the patient.	No. The antigen series is not a relevant patient series for the patient.	Yes. The antigen series is a relevant patient series for the patient.	No. The antigen series is not relevant patient series for the patient.

6 EVALUATE VACCINE DOSE ADMINISTERED

The core of a CDS engine is the process of evaluating a single vaccine dose administered against a defined target dose within a relevant patient series to determine if the vaccine dose administered is “valid” or “not valid” for the relevant patient series. The results will ultimately determine if all requirements of the target dose are satisfied. This can be accomplished by breaking the evaluation process into simple logical components. After processing each logical component, the results of those logical components are used to determine if the vaccine dose administered satisfies the goals of the target dose.

Each logical component has its own set of business rules that are used to determine if a target dose is “satisfied.” These business rules are documented using business rules and decision tables. (See section 2.11 to review an example of a decision table using a real-world scenario.) The decision table describes the way that the CDS engine responds to various combinations of conditions. The implementer can clearly see the set of conditions, how they work in combination, and what actions should be taken on a given set of conditions.

Specific attributes and decision tables are provided for each step of the evaluation process.

TABLE 6-1 EVALUATION PROCESS STEPS

Section	Activity	Goal
6.1	Evaluate Dose Administered Condition	The goal of this step is to determine if a vaccine dose administered can be evaluated.
6.2	Evaluate Conditional Skip	The goal of this step is to determine if the target dose can be skipped due to a patient’s age or immunization history.
6.3	Evaluate For Inadvertent Vaccine	The goal of this step is to determine if the vaccine dose administered was an inadvertent administration due to the vaccine type that was administered.
6.4	Evaluate Age	The goal of this step is to determine if the vaccine dose administered was given at an appropriate age.
6.5	Evaluate Preferable Interval	The goal of this step is to determine if the vaccine dose administered was given at an appropriate interval.
6.6	Evaluate Allowable Interval	The goal of this step is to determine if the vaccine dose administered was given at an allowable interval.
6.7	Evaluate Vaccine Conflict	The goal of this step is to determine if the vaccine dose administered was in conflict with any other vaccines.
6.8	Evaluate For Preferable Vaccine	The goal of this step is to determine if the vaccine dose administered was one of the preferable vaccines.
6.9	Evaluate For Allowable Vaccine	The goal of this step is to determine if the vaccine dose administered was one of the allowable vaccines.
6.10	Satisfy Target Dose	The goal of this step is to determine if the target dose is satisfied.